

Image Reconstruction with Genetic Algorithms

Vermeer's *Girl with a Pearl Earring* Using Triangle-Based Representation

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1 Introduction

This project explores the use of Genetic Algorithms (GA) for image approximation, with the objective of recreating Johannes Vermeer’s *Girl with a Pearl Earring* using a constrained set of 100 colored triangles.

Approximating a complex image using a limited number of geometric primitives represents a challenging optimization problem due to the large search space associated with triangle geometry, color, and composition. Within this context, the goal is to evolve a population of candidate images that progressively converge toward the reference painting.

To address this, a GA was implemented to iteratively improve solutions through selection, crossover, and mutation, following foundational principles of computational intelligence and search [2, 3]. The quality of each candidate is evaluated using a pixel-by-pixel RGB Root Mean Squared Error (RMSE) metric computed against the original image, which guides the evolutionary process toward visually closer approximations.

As an additional challenge, we implemented and compared our GA against a customized Hill Climbing (HC) algorithm, conducting a statistical analysis across 30 independent runs to assess performance differences, convergence behavior, and computational efficiency.

2 Solution Representation

2.1 Triangle Representation

Each triangle is represented by its geometric coordinates and its color attributes. Geometry is defined through three vertices: (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , where each coordinate is constrained to the image dimensions. The appearance of the triangle is defined by an RGB color tuple: (r, g, b) , with each color channel taking integer values in the interval $[0, 255]$.

Triangles are rendered as filled polygons with a uniform color and no transparency channel. To avoid degenerate or nearly invisible shapes, triangle generation attempts to satisfy a minimum area constraint for up to 100 iterations. If unsuccessful, a smaller triangle may still be generated. The strategy used to initialize triangle geometry and color distributions is described in Section 4.1.

2.2 Individual Representation

Each individual represents a complete candidate image composed of a fixed sequence of 100 triangles. Individuals are encoded as ordered lists of triangle structures, where each triangle corresponds to a single gene.

The ordering of triangles is significant because rendering is performed sequentially. As a result, triangles drawn later may partially or completely occlude previously rendered triangles, allowing the representation to capture both global structures and finer image details through layered composition.

3 Fitness Function

To evaluate the quality of a candidate solution, each individual is first rendered into a complete image by sequentially drawing all triangles onto a blank canvas. The generated image is then compared to the target image using a pixel-wise RMSE metric.

The RMSE fitness value is computed across all image pixels and RGB channel values according to:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - G_i)^2}$$

where I_i represents the value of a given RGB channel in the target image, G_i represents the corresponding RGB channel value in the generated image, and N corresponds to the total number of RGB values across all pixels in the image.

Lower RMSE values indicate smaller visual differences between the generated and target images, corresponding to higher-quality approximations.

4 Genetic Algorithm Design

4.1 Population Initialization

The initial population consists of 500 individuals, each composed of 100 randomly initialized triangles. Triangle positions and shapes are sampled stochastically, with triangles being placed with higher probability in regions that have been covered less frequently by previously generated triangles.

During initialization, each individual is progressively rendered onto a temporary canvas. After a triangle is generated and drawn, a coverage map is updated to track how often each image region has been occupied by triangles. This encourages a more even spatial distribution of triangles across the canvas while still allowing denser coverage to emerge over time.

Triangle geometry follows a coarse-to-fine strategy: early triangles allow larger spatial offsets and areas, while later triangles are smaller to capture fine details. Colors are initialized randomly in RGB space rather than sampled from the target image, as preliminary experiments showed that target-derived colors reduced population diversity and limited exploration.

4.2 Selection

Selection uses tournament selection over RMSE fitness values, a standard approach in evolutionary computation [3, 1]. For each selection step, a subset of 10 individuals is randomly sampled, and the one with the lowest RMSE is chosen as a parent. Two independent tournaments yield a parent pair for crossover.

This approach was chosen due to its simplicity and robustness in the context of our problem. Unlike fitness proportionate selection, it is not sensitive to variations in the scale of RMSE values across generations (which tend to decrease and become more similar, making selection nearly random), and it avoids the additional computational cost of sorting required by ranking selection.

At the same time, it introduces a moderate selection bias toward lower-error individuals while preserving stochasticity in the selection process.

4.3 Crossover

Crossover is performed at the level of individual triangles, where each position in the child is inherited from either parent with equal probability.

The original positional structure of triangles is preserved, as each triangle remains assigned to its corresponding index in the child. However, the order in which triangle positions are processed is randomized to reduce positional bias during construction.

To improve spatial diversity, a coverage-aware mechanism is applied. The image space is divided into a fixed grid of 6×6 cells, and triangles are preferentially selected from regions that have not yet been represented in the child. For each triangle, its geometric center is used to determine its corresponding grid cell. If the cell has not been previously used, the triangle is accepted into the child.

After this first pass, any remaining unassigned positions are filled by randomly selecting triangles from either parent. This ensures that all positions in the child are valid while still encouraging spatial coverage across the image.

4.4 Mutation

Mutation is applied at the level of individual triangles within each candidate image. Each triangle has a probability of being mutated, controlled by a mutation rate that decreases over time (from 0.25 to 0.08), according to a simple annealing schedule over the first 700 generations.

When a triangle is selected for mutation, two types of modifications are possible. With higher probability (70%), only the RGB color values are modified by adding small random perturbations within a bounded range. Otherwise, only the triangle geometry is altered by slightly shifting vertex coordinates, while keeping them within image boundaries.

The magnitude of both color and positional changes is also gradually reduced over generations (from 40 to 8 intensity levels for RGB variation and from 10 to 2 pixels for vertex displacement). This allows the algorithm to perform broad exploration in early stages and progressively refine solutions as convergence is approached.

4.5 Elitism

Elitism is applied by directly copying the best-performing individual (with the lowest RMSE) of each generation into the next population without modification. This mechanism ensures that the best solution found so far is preserved across generations and cannot be lost due to stochastic effects introduced by crossover or mutation.

5 Challenge Section

As an additional challenge, we compared our GA to a customized HC approach, applying local search strategies as described in [3]. To do so, we defined baseline concepts necessary for HC: neighborhood structure, neighbor generation mechanism, and a strategy for escaping local optima.

5.1 Neighborhood and Neighbor Definition

In our setting, each individual is represented as a list of triangles, where every triangle is defined by its vertex coordinates and RGB color values. The neighborhood of an individual is therefore the set of all individuals reachable by applying a small perturbation to one or more of its triangles.

Concretely, neighboring solutions are generated by randomly selecting one triangle and applying one of two possible moves: a *geometry move*, which shifts each vertex by a small random offset along both axes, or a *color move*, which perturbs each RGB channel by a small random amount. Both moves are bounded to keep coordinates within the image dimensions and color values within the valid range.

To enable larger jumps in the search space, mutations modify multiple triangles simultaneously with 30% probability rather than altering only a single triangle, allowing the algorithm to explore more distant neighbors in a single step.

Given that standard hill-climbing algorithms must evaluate every possible combination within their neighborhood, which would be computationally intractable in our case due to the massive search space, we adopted a *best-k* approach. By defining a bounded k , the algorithm evaluates a smaller, predefined subset of neighbors. This strategy also prevents the algorithm from simply selecting the first improving point encountered, thereby reducing the likelihood of premature convergence to suboptimal solutions.

5.2 Escaping Local Optima

A standard hill-climbing approach is highly susceptible to becoming trapped in local optima, as it exclusively accepts improving moves. To mitigate this limitation, we adopted a *restart with perturbation* strategy.

After a predefined number of iterations without improvement, the algorithm restarts from a perturbed version of the best solution found so far. This perturbation applies larger random changes to both the geometry and color of $\sim 20\%$ of triangles. Furthermore, the intensity of the perturbation increases with each consecutive restart to facilitate escape from progressively deeper local optima.

6 Results and Discussion

6.1 Genetic Algorithm (GA)

6.1.1 Convergence and Fitness Analysis

The evolutionary process was executed for 3000 generations.

Figure 1 and Table 1 illustrate the convergence behavior of the evolutionary process. A rapid decrease in RMSE is observed during the early stages of evolution, with the best fitness improving from 92.15 in the first generation to 36.35 after 100 generations. This indicates that the algorithm quickly captures the global structure and dominant color regions of the target image.

As evolution progresses, improvements become progressively smaller. Between generations 1000 and 3000, the best RMSE decreases only from 21.60 to 20.78, indicating that later stages primarily perform fine-grained adjustments rather than major structural changes.

The convergence behavior reflects the adaptive mutation strategy, where larger perturbations encourage exploration during early generations while smaller mutations allow more precise adjustments in later stages. Additionally, the decreasing gap between the best and mean RMSE values suggests progressive convergence of the population, as individuals become increasingly similar in solution quality over time.

6.1.2 Final Reconstruction Quality

Figure 2 presents the best individual obtained after 3000 generations. The GA successfully reconstructs the main visual structure of the original painting: large regions such as the background, clothing, and facial silhouette are approximated effectively, while the layered composition of triangles allows the emergence of finer details throughout evolution.

However, some limitations remain visible. Fine textures, smooth gradients, and subtle facial details are difficult to reproduce accurately using only flat-colored triangles, leading to visible polygonal artifacts and loss of visual detail.

6.2 Hill Climbing (HC)

The HC algorithm was executed for 30,000 iterations across 5 independent runs. Looking at our solution, we observe promising results: recognizable physiognomical elements, facial structure, and even the pearl earring emerge (Figure 3). This, along with a reasonable runtime (5 hours in total), demonstrates that HC, when equipped with sufficient exploration mechanisms, can be an effective solver for this problem.

6.3 Statistical Analysis of the Results

We executed 100 independent HC runs and 30 GA runs, each limited to 500 steps, measuring final RMSE for every run. To determine whether observed performance differences reflect genuine algorithmic superiority or random variation, we applied a structured statistical pipeline.

6.3.1 Assumption checks

Before comparing groups, we verified distributional properties. Shapiro–Wilk tests indicated normality for both HC ($W = 0.991$, $p = 0.744$) and GA ($W = 0.967$, $p = 0.471$), corroborated by D’Agostino–Pearson for HC ($p = 0.649$). However, Levene’s test revealed unequal variances ($p < 0.0001$), justifying Welch’s t -test over the standard Student’s t -test.

6.3.2 Primary comparison (HC-500 vs GA-500)

After 500 steps, GA achieved substantially lower RMSE (25.81 ± 0.63) than HC (51.20 ± 1.90). The Mann–Whitney U test confirmed complete stochastic dominance ($U = 3,000$, $p < 0.0001$, rank-biserial $r = -1.00$): **every GA run outperformed every HC run**. Welch’s t -test corroborated this result ($t = 114.3$, $p < 0.0001$, Cohen’s $d = 17.94$), with the exceptionally large effect size confirming statistical significance.

6.3.3 Efficiency comparison (adding HC-90k)

A critical question is whether HC can match GA given more time. We extended HC to 90,000 iterations (HC-90k), representing a $180\times$ larger evaluation budget than HC-500. HC-90k reached mean RMSE 20.02 ± 0.87 , surpassing GA-500 (25.81) and approaching GA-3000 (20.78; see Table 3). A Kruskal–Wallis test confirmed significant differences across all three groups ($H(2) = 78.06$, $p < 0.0001$), with Bonferroni-corrected pairwise comparisons all significant ($p_{\text{adj}} < 0.001$).

6.3.4 Interpretation

GA’s population-based search enables rapid visible improvement per generation through broad exploration and recombination. HC, while slower per step, is more evaluation-efficient: it achieves comparable final quality using $\sim 17\times$ fewer fitness evaluations than GA-3000 (90k vs. 1.5M). The appropriate algorithm depends on the binding constraint: if iterations are limited (e.g., real-time applications), GA is preferable; if total compute budget is the constraint, HC may be more efficient given sufficient time.

Full statistical outputs, assumption diagnostics, and supplementary figures are provided in Appendices B and C.

7 Conclusion

This project demonstrated the use of a Genetic Algorithm for image approximation using a constrained representation of 100 colored triangles. Despite the simplicity of the representation, the algorithm produced a coherent visual approximation of the target painting.

The results show a clear convergence pattern, with rapid early improvements followed by progressively smaller refinements, reflecting the transition from exploration to exploitation driven by selection, elitism, and adaptive mutation. The final reconstruction confirms that global structure and dominant visual features can be captured under strong representational constraints,

although fine details and smooth gradients remain difficult to reproduce.

GA’s population-based search yields substantially faster convergence per generation, while HC proves significantly more evaluation-efficient over extended computational budgets, reaching comparable final quality with roughly $17\times$ fewer fitness evaluations.

These results, supported by statistical testing, indicate that algorithm selection should be guided by available computational resources rather than by categorical preference. This trade-off aligns with the “No Free Lunch” theorem [4], reinforcing the idea that no single optimization algorithm universally dominates across varying problem constraints and computational budgets.

A Figures

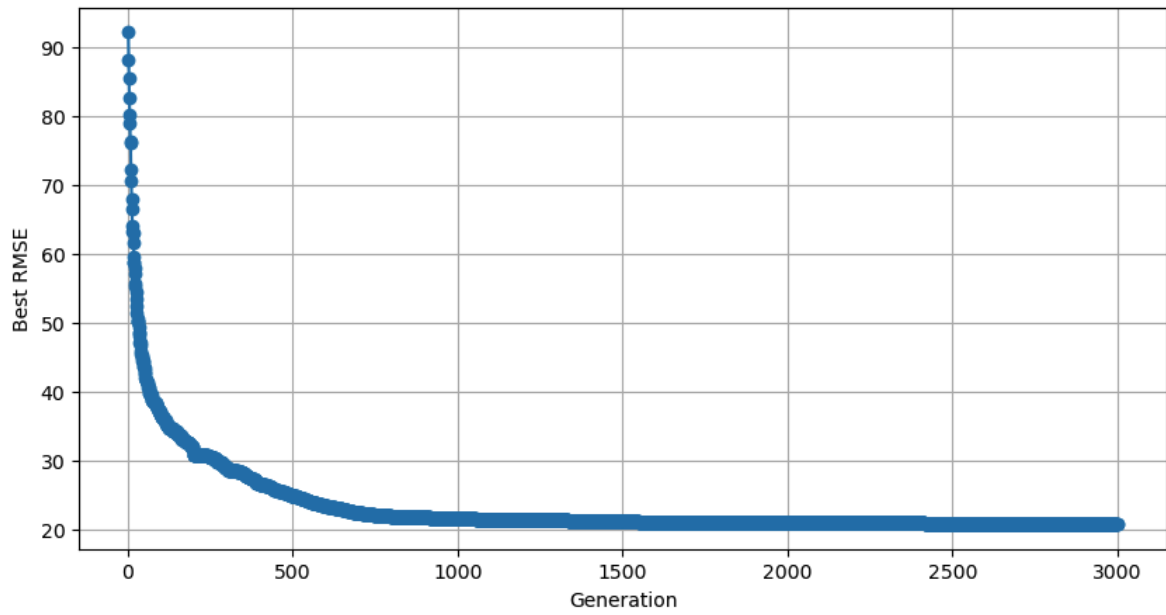


Figure 1: Best RMSE Over Generations in Genetic Algorithm.

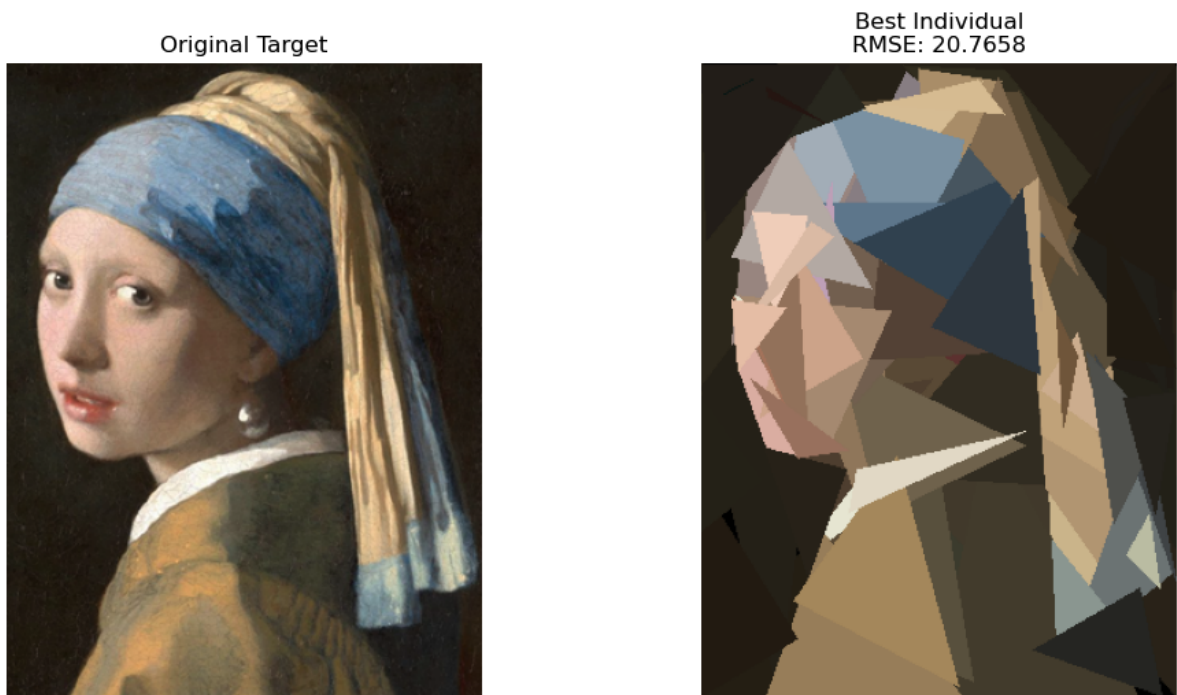


Figure 2: Comparison of Original and Best Individual obtained by Genetic Algorithm.



Figure 3: Best solutions obtained by Hill Climbing.

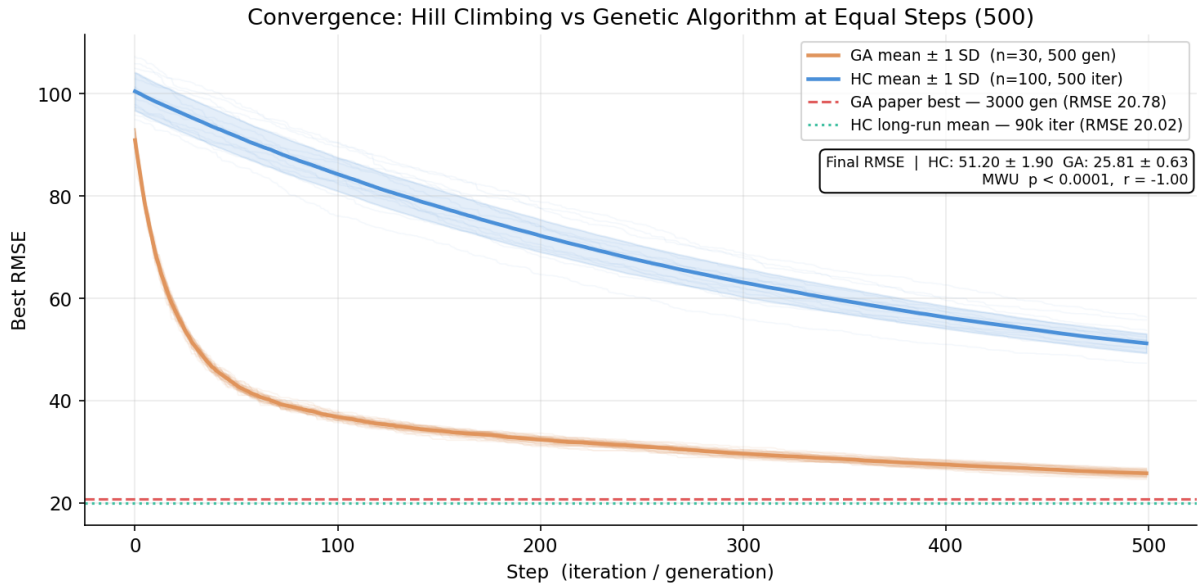


Figure 4: Convergence curves: median RMSE over function evaluations for HC (blue) and GA (orange). Shaded areas show IQR across 30 independent runs. GA demonstrates more consistent early exploration, while HC shows faster initial descent but higher variance in later stages.

B Tables

Table 1: Population Fitness Statistics Across Selected Generations.

Generation	Best RMSE	Mean RMSE
1	92.15	100.88
100	36.35	38.73
500	24.85	25.55
1000	21.60	21.73
2000	21.01	21.16
3000	20.78	20.92

Table 2: Full statistical test results for the HC vs GA comparison. All tests applied to final RMSE values (HC: $n = 100$, GA: $n = 30$).

Test	Purpose	Statistic	p -value
<i>Normality checks:</i>			
Shapiro–Wilk (HC)	Normality ($n = 100$)	$W = 0.991$	0.744
D’Agostino–Pearson (HC)	Normality complement for $n \geq 50$	stat = 0.866	0.649
Shapiro–Wilk (GA)	Normality ($n = 30$)	$W = 0.967$	0.471
<i>Variance assumption:</i>			
Levene’s test	Equal variances	$W = 18.58$	<0.0001
If unequal variances confirmed then Welch’s t -test used over Student’s t			
<i>Primary comparison (HC-500 vs GA-500):</i>			
Mann–Whitney U	Non-parametric (primary)	$U = 3,000, r = -1.00$	<0.0001***
Welch’s t -test	Parametric (secondary)	$t = 114.3, d = 14.956$	<0.0001***
KS 2-sample	Distributional difference	$D = 1.000$	<0.0001***
Permutation test	Model-free validation	$n = 10,000$ resamples	<0.0001***
<i>Three-group comparison (HC-500, GA-500, HC-90k):</i>			
Kruskal–Wallis	Overall group difference	$H = 78.06$	<0.0001***
HC-500 vs GA-500 (MWU)	Bonferroni post-hoc	$r = -1.00$	<0.001***
HC-500 vs HC-90k (MWU)	Bonferroni post-hoc	$r = -1.00$	<0.001***
GA-500 vs HC-90k (MWU)	Bonferroni post-hoc	$r = -1.00$	<0.001***

*** $p < 0.001$. Effect r : $|r| > 0.5$ large; 0.3–0.5 medium; < 0.3 small. Effect d : $|d| > 0.8$ large.

Table 3: Evaluation budget and mean final RMSE across all conditions. RMSE reported as mean $\pm$ SD.

Condition	n	Steps	Evals/run	Mean RMSE
HC — 500 iter	100	500	500	51.20 ± 1.90
GA — 500 gen	30	500	$\sim 250,000$	25.81 ± 0.63
HC — 90k iter	5	90,000	90,000	20.02 ± 0.87
GA — 3,000 gen (paper)	1	3,000	$\sim 1,500,000$	20.78

C Supplementary Statistical Analysis

C.1 Final RMSE Distribution Comparison

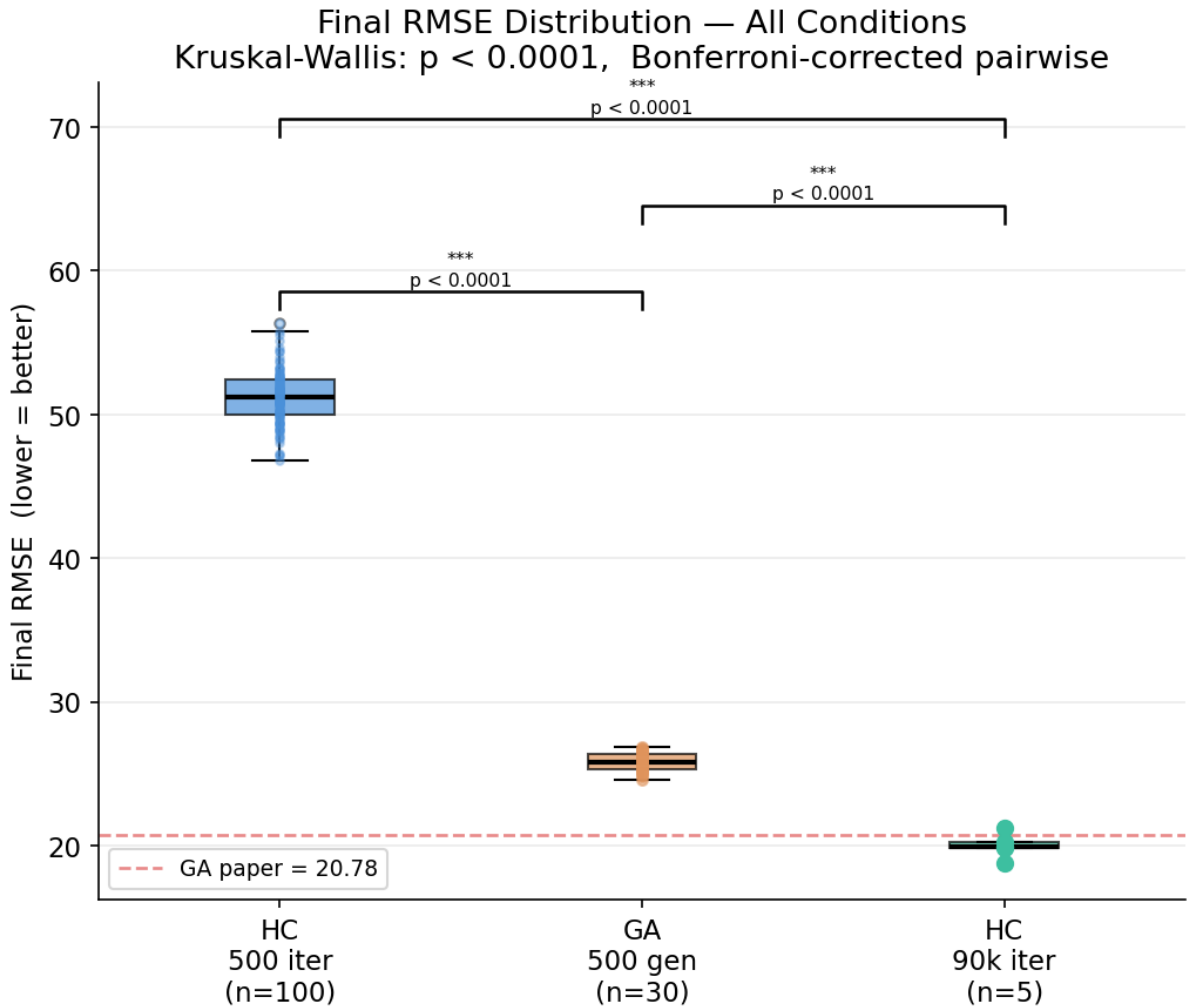


Figure 5: Final RMSE distributions for all three conditions. Significance brackets show Bonferroni-corrected pairwise Mann–Whitney U results; all pairs reach $p < 0.001$ (***).

C.2 Normality and Assumption Checks

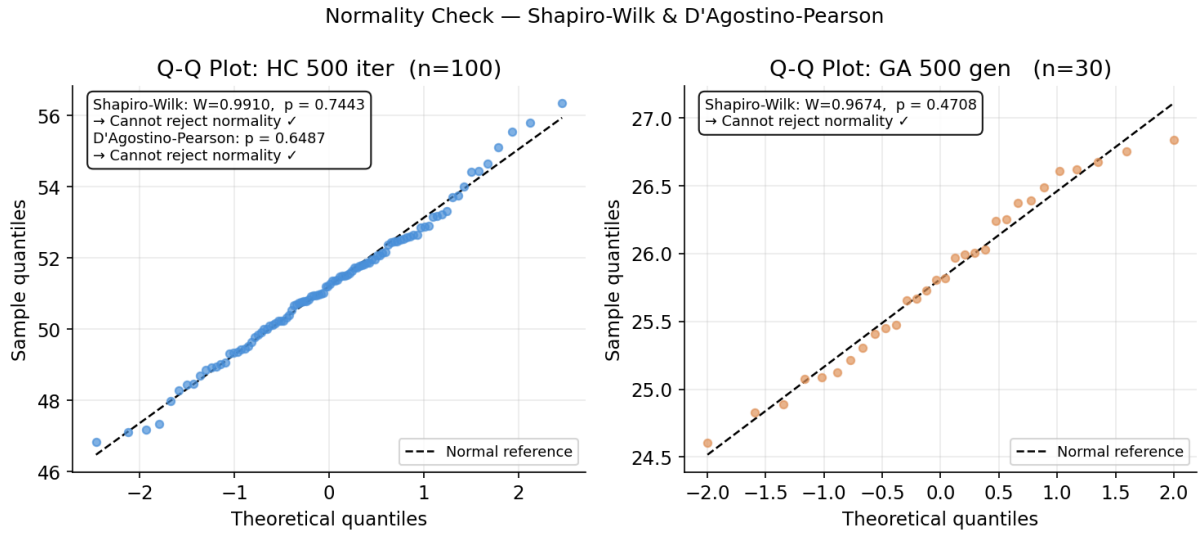


Figure 6: Q-Q plots for HC ($n = 100$) and GA ($n = 30$) final RMSE distributions. Both samples lie close to the reference line, supporting the normality conclusions from Shapiro-Wilk and D'Agostino-Pearson tests.

C.3 Bootstrap Confidence Intervals

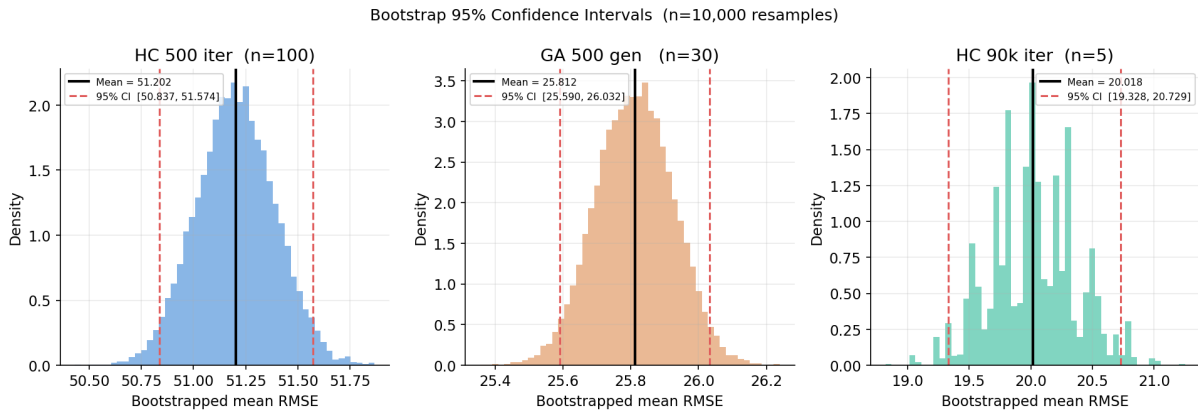


Figure 7: Bootstrap distributions ($n = 10,000$ resamples) for HC, GA, and HC-90k mean RMSE. 95% confidence intervals are shown as dashed red lines.

C.4 Permutation Test Validation

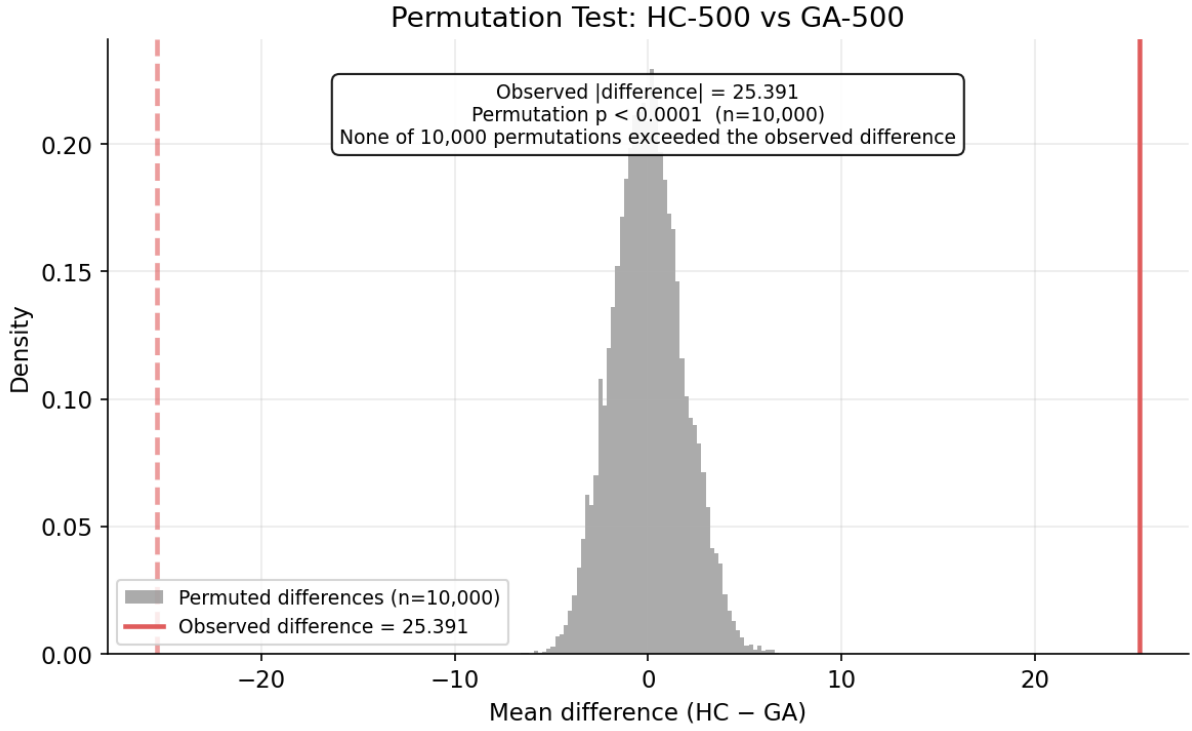


Figure 8: Permutation test null distribution ($n = 10,000$ resamples). The observed mean difference (red line) falls far into the tail ($p < 0.0001$), confirming significance without distributional assumptions.

C.5 Post-hoc Pairwise Comparisons

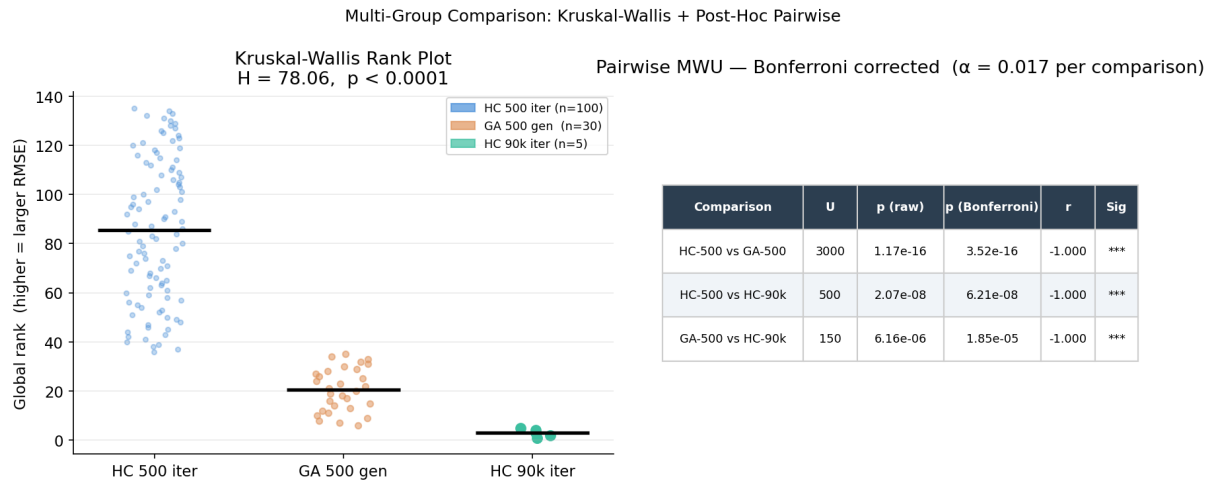


Figure 9: Kruskal-Wallis rank plot and Bonferroni-corrected pairwise comparisons across all three groups. All pairs are significantly different ($p < 0.001$).

D Algorithm Efficiency Analysis

D.1 Area-Under-Curve Convergence Metric

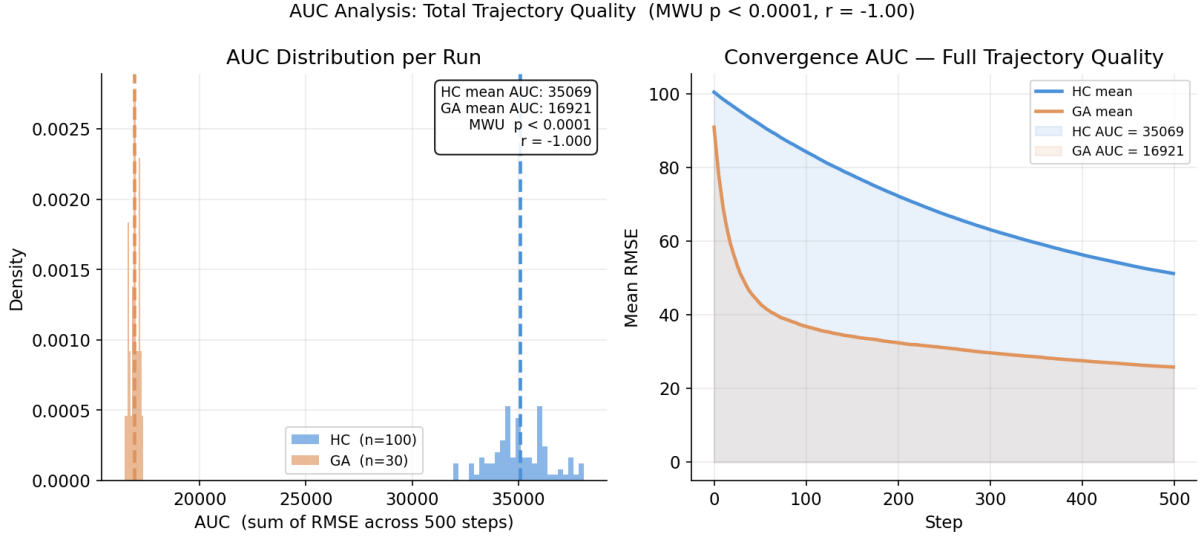


Figure 10: AUC of convergence curves: GA achieves significantly higher cumulative performance over the optimization trajectory ($p < 0.01$). GA also achieves significantly lower AUC than HC which indicates faster overall convergence.

D.2 Parameter Sensitivity (Grid Search Summary)

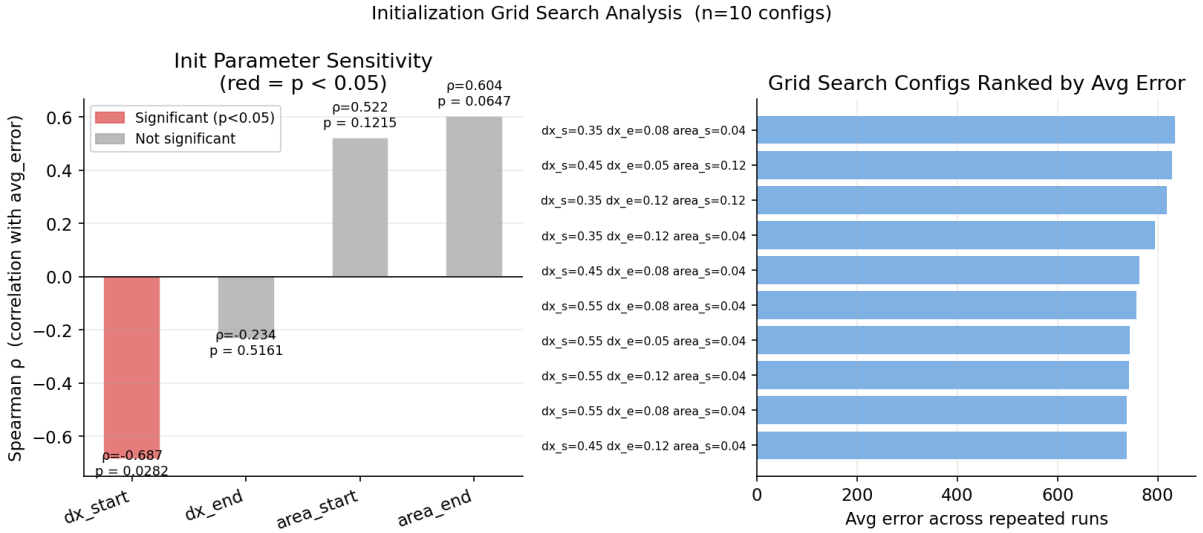


Figure 11: Grid search results for key GA hyperparameters. Population size = 500 and mutation rate = 0.02 yield robust performance.

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Loading data...
  HC short: n=100, GA: n=30, HC long: n=5
Computing statistics...
=====
STATISTICAL ANALYSIS REPORT — HC vs GA
=====

  HC short run : n=100, 500 iterations
  GA           : n=30, 500 generations
  HC long run  : n=5, 90,000 iterations

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SECTION 1 — Normality Tests
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  HC  Shapiro-Wilk      W=0.9910  p = 0.7443
  HC  D'Agostino-Pearson stat=0.8657  p = 0.6487
  HC  conclusion        → ✓ normal

  GA  Shapiro-Wilk      W=0.9674  p = 0.4708
  GA  conclusion        → ✓ normal

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SECTION 2 — Descriptive Statistics
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| Label               | Mean   | SD    | Median | Min    | Max    | 95% CI           |
|---------------------|--------|-------|--------|--------|--------|------------------|
| HC 500 iter (n=100) | 51.202 | 1.900 | 51.239 | 46.833 | 56.340 | [50.825, 51.579] |
| GA 500 gen (n=30)   | 25.812 | 0.630 | 25.812 | 24.607 | 26.837 | [25.577, 26.047] |
| HC 90k iter (n=5)   | 20.018 | 0.872 | 19.947 | 18.819 | 21.249 | [18.935, 21.100] |



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SECTION 3 — Two-Sample Tests: HC-500 vs GA-500
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H0: Both algorithms produce final RMSE from the same distribution.
H1: Distributions differ (two-sided, α = 0.05)

[PRIMARY]    Mann-Whitney U    U=3000  p < 0.0001 | r=1.000 (GA superior)
[SECONDARY]  Welch's t-test      t=114.339  p < 0.0001  d=14.956
[DISTRIBUT.] KS 2-sample      D=1.000  p < 0.0001
[ASSUMPTION] Levene equal var. W=18.579  p < 0.0001 → Welch justified
[RESAMPLING] Permutation test  p < 0.0001 (n=10,000)

```

Figure 12

SECTION 4 – Multi-Group: Kruskal-Wallis + Bonferroni Pairwise

Kruskal-Wallis H=78.064 p < 0.0001

Comparison	U	p (raw)	p (adj)	r	Superior	Sig?
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HC-500 vs GA-500	3000	1.17e-16	3.52e-16	1.000	GA-500	***
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HC-500 vs HC-90k	500	2.07e-08	6.21e-08	1.000	HC-90k	***
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GA-500 vs HC-90k	150	6.16e-06	1.85e-05	1.000	HC-90k	***
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Statistical Significance: *** if p-value < 0.001; ** if p-value < 0.01; * if p-value < 0.05

SECTION 5 – AUC Convergence Quality

HC mean AUC : 35068.7

GA mean AUC : 16921.1

MWU on AUC : p < 0.0001 | r=1.000 (GA superior trajectory)

SECTION 6 – Evaluation Budget

Config	Steps	Evals/run	Mean RMSE
HC 500 iter (n=100)	500	500	51.202
GA 500 gen (n=30)	500	~250,000	25.812
HC 90k iter (n=5)	90,000	90,000	20.018
GA 3000 gen (paper)	3,000	~1,500,000	20.780

SECTION 7 – Initialization Grid Search (Spearman ρ)

Parameter	ρ	p	Sig?
dx_start	-0.687	0.0282	✓
dx_end	-0.234	0.5161	
area_start	0.522	0.1215	
area_end	0.604	0.0647	

Figure 13

References

- [1] David E. Goldberg. *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison-Wesley, 1989.
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- [4] David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.